Cross-Hospital Generalization of Heart-Disease Classifiers

A Directed Transfer Analysis on Four UCI Cohorts

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A clinical risk model can appear excellent under internal validation and still fail when it is moved to a hospital with a different case mix, different measurement practice, and a different missing-data pattern. We quantify that failure mode on the four cohorts of the UCI Heart Disease collection (Cleveland, Hungary, Switzerland, Long Beach VA; n = 920 patients). Three model families—L_2-penalized logistic regression, a random forest, and a compact multilayer perceptron—are first evaluated on Cleveland with stratified five-fold cross-validation using all thirteen recorded variables, and are then transported across all twelve directed source→target hospital pairs using only the nine variables with genuine coverage at every site. Imputation and scaling are refitted strictly inside each training boundary, and the Swiss cholesterol field—recorded as zero for every Swiss patient—is excluded from the portable feature set rather than silently imputed from other hospitals. Internal Cleveland discrimination reaches AUC = 0.910 for the random forest. Under transport, however, the picture is not a uniform degradation but a heterogeneous matrix: individual directed pairs range from AUC = 0.571 to 0.888, and the random forest attains a *higher* macro-average external AUC (0.781) than internal AUC (0.756), while logistic regression loses 0.028 AUC and the MLP is essentially flat but weaker in absolute terms. We argue that the scalar "generalization gap" routinely reported in the clinical machine-learning literature is an insufficient summary, that source-cohort quality dominates model choice, and that external validity should be reported as a transfer matrix accompanied by absolute target-site performance.

CCS Concepts: • **Applied computing** → **Health informatics**; • **Computing methodologies** → *Supervised learning by classification*; *Cross-validation.*

Additional Key Words and Phrases: Clinical machine learning, dataset shift, external validation, heart disease, ROC analysis, reproducibility, transportability.

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1 INTRODUCTION

The standard development pipeline for a clinical prediction model ends with a held-out estimate of discrimination computed on patients drawn from the same institution that supplied the training data. That estimate answers a narrow question: how well does the model rank patients *within the population it was fitted on*? The deployment question is different. When a model is installed at a second hospital, three things change simultaneously—the covariate distribution $P(X)$, the outcome prevalence $P(Y)$, and the measurement and missingness process that maps patients to recorded variables. Each can degrade performance independently, and the literature contains repeated demonstrations that internal validation is a poor predictor of external behaviour [5, 9, 12].

The UCI Heart Disease collection [3] is unusually well suited to studying this because it is not one dataset but four, each collected at a different institution with a shared instrument but very different recording discipline. It is also small, historical, and imperfect, which makes it a good teaching object and a bad basis for clinical claims. We use it accordingly.

1.1 Contributions

(1) We report the full 4×4 directed transfer matrix of ROC-AUC for three model families, rather than a single pooled external score, and show that the matrix is strongly asymmetric: transfer quality is governed more by the *source* cohort than by the model.
(2) We formalise the routinely reported "generalization gap" as a signed contrast between two macro-averages over different estimands, and demonstrate a concrete case (the random forest) in which the gap is negative—external exceeds internal—so that the statistic cannot be read as a robustness score.
(3) We identify and correct a leakage pattern specific to this benchmark: the Swiss cholesterol column is identically zero, and treating it as an observed shared feature imports information from other hospitals into a variable the source site never measured.
(4) We supply a fully specified, seed-fixed protocol in which every adaptive operation is refitted inside the training boundary, and release the pairwise score table so that the conclusions can be re-derived without re-running the models.

2 DATA

2.1 Cohorts

The collection contains 920 patients across four institutions, summarised in Table 1. Each record carries thirteen candidate predictors and a diagnosis code $\texttt{num} \in \{0, 1, 2, 3, 4\}$ indicating angiographic disease severity. Following standard practice for this benchmark we binarise the outcome,

$$y_i = \mathbb{1}\left[\texttt{num}_i > 0\right], \tag{1}$$

so that $y_i = 1$ denotes the presence of coronary artery disease.

[1]The study uses the public UCI Heart Disease collection and is intended as methodological research on external validation. It is not a diagnostic device and must not be used for clinical decisions.

Table 1. The four source institutions. Prevalence is the fraction of patients with `num` > 0. "Absent variables" counts predictors that the site never records or records as a constant sentinel.

Institution	Patients	Prevalence	Notable recording defects
Cleveland	303	0.459	4 missing `ca`, 2 missing `thal`
Hungary	294	0.361	`ca` (291), `thal` (266), `slope` (190) absent
Switzerland	123	0.935	`chol` = 0 for *all* 123 patients; 8 negatives
Long Beach VA	200	0.745	56 missing `trestbps`, 49 zero `chol`
Total	920	0.553	—

2.2 Disguised missing values

Two variables admit physiologically impossible zeros. Serum cholesterol (`chol`) and resting blood pressure (`trestbps`) cannot be zero in a living patient, so any recorded zero is a sentinel for a missing measurement. We map these to `NaN` before any other processing. The consequence is severe at one site: Switzerland records `chol = 0` for all 123 patients, meaning the variable is not merely sparse but entirely unobserved there.

This matters for the transport experiment. If cholesterol is retained in the shared feature set and imputed with a median computed from the pooled data, then a model "trained on Switzerland" receives, for every Swiss patient, a cholesterol value manufactured from other hospitals. The model then appears to transport a variable that its source site never measured. We therefore exclude `chol` from the portable set entirely—a stricter and more defensible choice.

2.3 Feature sets

Two disjoint estimands are studied, and they use different inputs.

- $\mathcal{F}_{13}$ (internal study): all thirteen predictors, including `ca`, `thal`, and `slope`, which are among the strongest signals but are recorded reliably only at Cleveland.
- $\mathcal{F}_9$ (transport study): `age`, `sex`, `cp`, `trestbps`, `fbs`, `restecg`, `thalach`, `exang`, `oldpeak`—the nine variables with genuine coverage at all four sites.

Scores computed on $\mathcal{F}_{13}$ and $\mathcal{F}_9$ answer different questions and are *not* comparable as if only the hospital had changed. We keep them separate throughout.

3 METHODS

3.1 Preprocessing inside the training boundary

Let $\mathcal{D}_{\text{tr}}$ denote the rows a model is permitted to learn from. We compute the column-wise median $\mathbf{m}$ and the standardisation statistics $(\boldsymbol{\mu}, \boldsymbol{\sigma})$ using $\mathcal{D}_{\text{tr}}$ only,

$$m_j = \text{median}\left\{x_{ij} : i \in \mathcal{D}_{\text{tr}},\ x_{ij} \text{ observed}\right\}, \tag{2}$$

and apply the fitted transform

$$\tilde{x}_{ij} = \frac{\left(x_{ij} \text{ if observed else } m_j\right) - \mu_j}{\sigma_j} \tag{3}$$

to both training and evaluation rows. In the internal study $\mathcal{D}_{\text{tr}}$ is the four training folds; in the transport study it is the entire source hospital. No statistic is ever computed from evaluation rows.

3.2 Models

3.2.1 Logistic regression. The linear baseline models the log-odds of disease as

$$\log \frac{\Pr(y = 1 \mid \tilde{\mathbf{x}})}{\Pr(y = 0 \mid \tilde{\mathbf{x}})} = \beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}, \tag{4}$$

with parameters obtained by minimising the class-weighted penalised negative log-likelihood

$$\mathcal{L}(\boldsymbol{\beta}) = -\sum_i w_{y_i} [y_i \log p_i + (1 - y_i) \log(1 - p_i)] + \frac{1}{2C} \|\boldsymbol{\beta}\|_2^2, \tag{5}$$

where $p_i = \sigma(\beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_i)$ and the balancing weights

$$w_c = \frac{n}{2\, n_c} \tag{6}$$

counteract prevalence imbalance. For the internal study C is chosen from $\{0.05, 0.1, 0.3, 1.0\}$ by an *inner* five-fold cross-validation nested within each outer training split; the transport study fixes $C = 1$ so that no target information can influence tuning.

3.2.2 Random forest. The ensemble averages $B = 400$ (internal) or $B = 300$ (transport) CART trees, each grown on a bootstrap resample with $\lfloor \sqrt{p} \rfloor$ candidate features per split and a minimum of three samples per leaf [1]:

$$\hat{p}(\tilde{\mathbf{x}}) = \frac{1}{B} \sum_{b=1}^{B} \hat{p}_b(\tilde{\mathbf{x}}). \tag{7}$$

3.2.3 Multilayer perceptron. A two-hidden-layer network with widths $(32, 16)$ and ReLU activations,

$$\hat{p}(\tilde{\mathbf{x}}) = \sigma\big(\mathbf{w}_3^\top \phi(\mathbf{W}_2\, \phi(\mathbf{W}_1 \tilde{\mathbf{x}} + \mathbf{b}_1) + \mathbf{b}_2) + b_3\big), \tag{8}$$

is trained with Adam, L_2 penalty $\alpha = 10^{-2}$, and early stopping on a 20% internal validation split with patience 20. It is a compact benchmark, not a clinical deep-learning system.

3.3 Evaluation protocol

3.3.1 Internal validation. Cleveland is partitioned into five stratified folds. For each fold the preprocessing of Equation (3) and the model are refitted on the other four, giving every patient exactly one out-of-fold score $\hat{p}_i$. Discrimination is the ROC-AUC of the pooled out-of-fold scores [4, 6],

$$\mathrm{AUC} = \frac{1}{n_+ n_-} \sum_{i:y_i=1} \sum_{j:y_j=0} \left[\mathbb{1}(\hat{p}_i > \hat{p}_j) + \tfrac{1}{2} \mathbb{1}(\hat{p}_i = \hat{p}_j) \right], \tag{9}$$

which is the probability that a randomly chosen diseased patient is ranked above a randomly chosen healthy one. Accuracy and F_1 at the fixed threshold 0.5 are reported as supporting endpoints only.

3.3.2 Directed transport. Let $\mathcal{S} = \{\text{Cleveland}, \text{Hungary}, \text{Switzerland}, \text{VA}\}$. For every ordered pair (s, t) with $s \neq t$ we fit preprocessing and model on the whole of site s and score all patients of site t, giving $\mathrm{AUC}_{s \to t}$. The diagonal entries $\mathrm{AUC}_{s \to s}$ are computed by five-fold cross-validation *within* site s, with preprocessing refitted per fold. This distinction is essential: without it, diagonal entries would be optimistically biased by statistics leaked from the held-out fold.

Table 2. Internal validation on Cleveland, stratified five-fold cross-validation, all thirteen predictors ($\mathcal{F}_{13}$). Threshold-dependent metrics use $\hat{p} \geq 0.5$.

Model	Accuracy	ROC-AUC	F_1
Logistic regression	0.838	0.907	0.821
Random forest	**0.842**	**0.910**	**0.821**
MLP	0.779	0.883	0.745

3.3.3 The generalization gap. We define the macro-averages

$$\overline{A}_{\text{int}} = \frac{1}{4}\sum_{s\in\mathcal{S}} \text{AUC}_{s\to s}, \qquad \overline{A}_{\text{ext}} = \frac{1}{12}\sum_{s\neq t} \text{AUC}_{s\to t}, \tag{10}$$

and the signed gap

$$\Delta = \overline{A}_{\text{int}} - \overline{A}_{\text{ext}}. \tag{11}$$

We stress that Equation (11) is a *descriptive contrast between two different estimands*, not a robustness statistic: the internal and external averages are taken over different test populations of different difficulty, so $\Delta < 0$ does not mean a model improves under shift, and $\Delta \approx 0$ does not mean a model is robust.

3.4 Reproducibility

All randomness is seeded at 42. Preprocessing, hyperparameter grids, feature lists, and the twelve directed scores are exported. The three model families are implemented in scikit-learn [7] and the full study runs on a laptop CPU in under ten seconds.

4 RESULTS

4.1 Internal discrimination on Cleveland

Table 2 and Figure 1 give the five-fold cross-validated performance on Cleveland using all thirteen variables. The random forest and logistic regression are statistically indistinguishable at this sample size (AUC = 0.910 versus 0.907); the MLP is clearly weaker (AUC = 0.883), consistent with the expectation that a 303-patient tabular problem does not support a flexible neural estimator [8].

Taken alone, Table 2 would support a confident claim: an AUC $\approx$ 0.91 classifier for coronary artery disease. The remainder of the paper is an argument that this claim is not yet a statement about anything outside Cleveland.

4.2 The transfer matrix

Figure 2 shows the full 4×4 directed matrix per model, and Table 3 lists the twelve off-diagonal scores. Three structural facts stand out.

Transfer is not uniformly downward. Of the twelve directed pairs, several *exceed* the corresponding internal score. Cleveland→Hungary reaches 0.875–0.878, above Cleveland's own within-site value of 0.861–0.870 on the same nine features. Hungary is an easier target: its prevalence is the lowest of the four (0.361) and its recorded variables are comparatively clean. A pooled "external AUC" that averages Hungary together with Switzerland conceals a factor-of-two difference in difficulty.

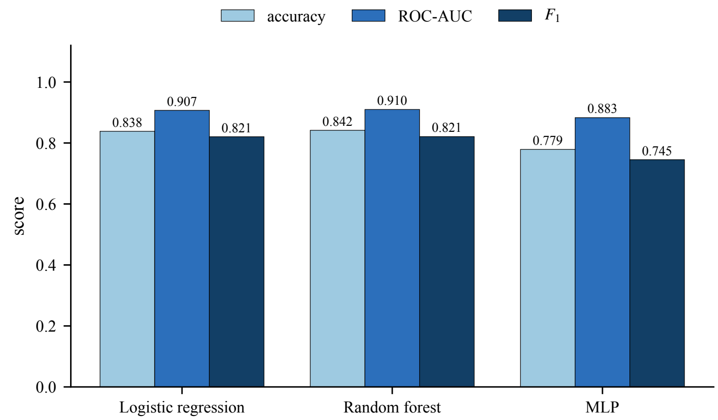

Fig. 1. Internal Cleveland performance under stratified five-fold cross-validation with all thirteen predictors. The two classical learners are separated by less than one standard error of the AUC at this sample size; the neural network is not competitive.

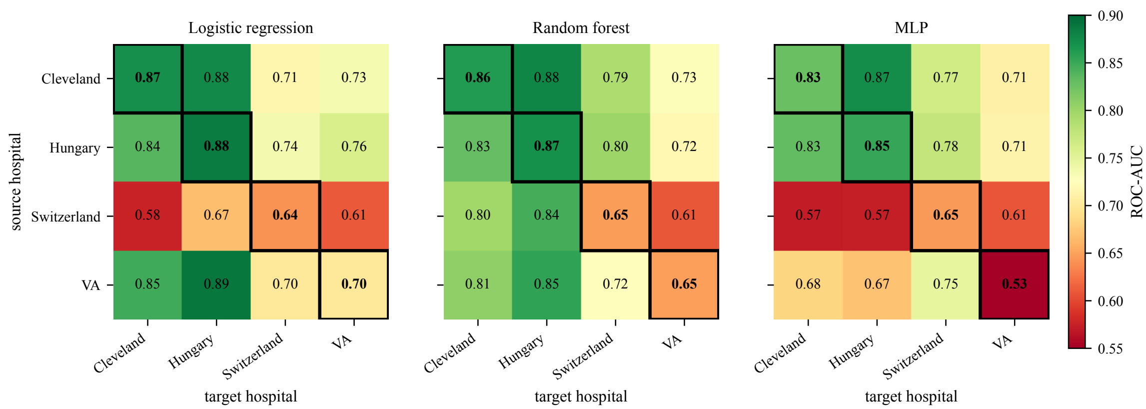

Fig. 2. Directed transfer matrices of ROC-AUC on the nine portable variables ($\mathcal{F}_9$). Rows index the source hospital used for fitting, columns the target hospital used for scoring. Boxed diagonal cells are within-site five-fold cross-validation with preprocessing refitted per fold; all other cells are strict source-only fits scored on an untouched target. The dominant structure is horizontal (source-driven), not vertical.

The source cohort dominates. Reading Figure 2 by rows rather than columns is more informative. Switzerland as a *source* is catastrophic for the two linear-ish learners: with 93.5% of Swiss patients diseased, the fitted logistic model sees almost no negative examples and transfers at AUC = 0.576 and 0.667 to Cleveland and Hungary— barely above chance. The random forest, whose bootstrap resampling and balanced class weights make it far less sensitive to an extreme prevalence, recovers 0.797 and 0.844 from the same degenerate source. Conversely, Switzerland as a *target* is uniformly hard for all models (0.700–0.799), because its own outcome distribution leaves only eight negative patients to rank against.

Model ranking is not stable across pairs. Logistic regression wins five of twelve pairs, the random forest six, the MLP one. No model dominates. A practitioner selecting a model from a single held-out target would reach a different

Table 3. All twelve directed source⟶target transfers on $\mathcal{F}_9$. Best model per row in bold.

Source	Target	Logistic	Random forest	MLP
Cleveland	Hungary	0.875	**0.878**	0.872
Cleveland	Switzerland	0.714	**0.789**	0.765
Cleveland	VA	**0.734**	0.728	0.709
Hungary	Cleveland	**0.836**	0.830	0.832
Hungary	Switzerland	0.736	**0.799**	0.775
Hungary	VA	**0.760**	0.718	0.713
Switzerland	Cleveland	0.576	**0.797**	0.571
Switzerland	Hungary	0.667	**0.844**	0.572
Switzerland	VA	**0.609**	0.608	0.606
VA	Cleveland	**0.848**	0.806	0.683
VA	Hungary	**0.888**	0.854	0.670
VA	Switzerland	0.700	0.723	**0.746**
Macro-average external		0.745	**0.781**	0.709

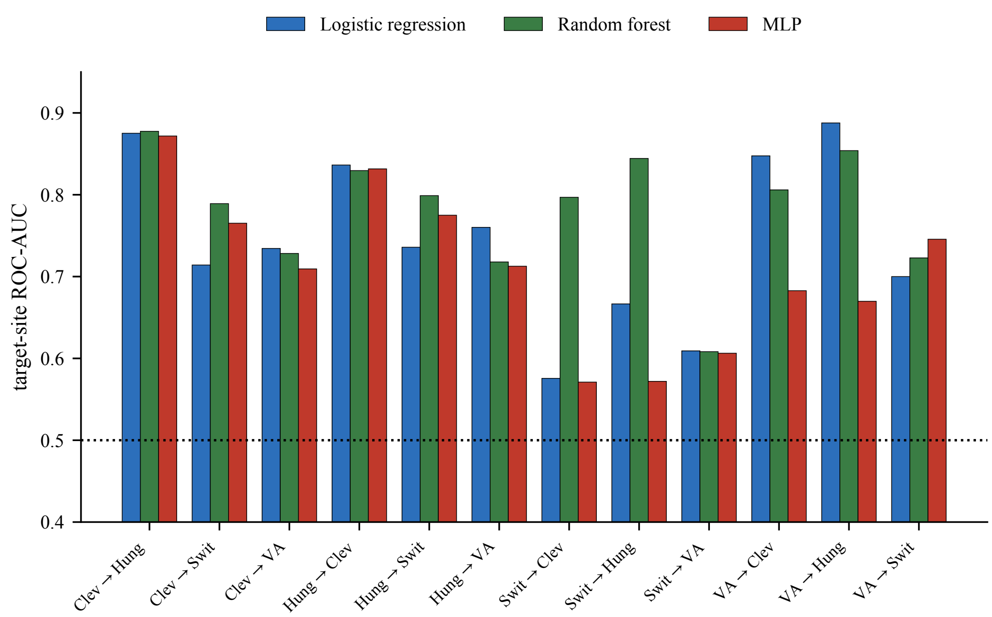

Fig. 3. The twelve directed transfers, grouped by source→target pair. The spread *within* a pair (model choice) is generally smaller than the spread *across* pairs (cohort choice), except where the source cohort is degenerate (Switzerland), in which case model choice becomes decisive.

conclusion depending on which target they happened to hold out—which is precisely the failure mode that motivates reporting the whole matrix.

Table 4. Macro-averaged internal and external discrimination on $\mathcal{F}_9$, and the signed gap of Equation (11). A negative gap does not indicate improvement under shift; see Section 5.

Model	$\overline{A}_{\text{int}}$	$\overline{A}_{\text{ext}}$	Δ
Logistic regression	0.774	0.745	+0.028
Random forest	0.756	**0.781**	−0.025
MLP	0.713	0.709	+0.003

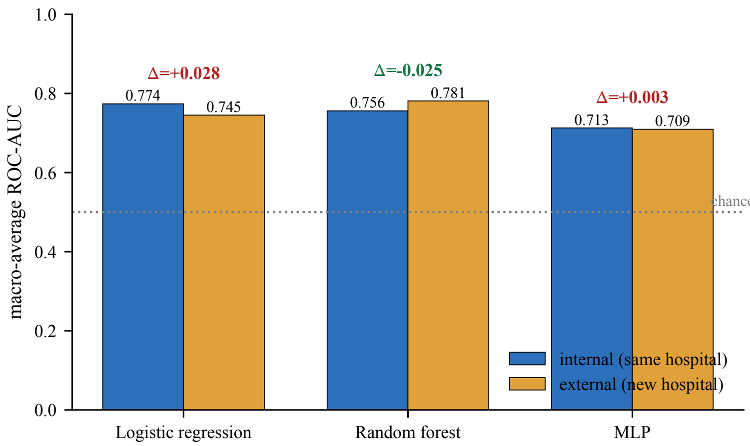

Fig. 4. Macro-averaged internal versus external discrimination. The MLP has the smallest magnitude gap and the worst absolute performance; the random forest has a negative gap and the best absolute external performance. Ranking models by $|\Delta|$ would select the weakest model.

4.3 The generalization gap is not a robustness score

Table 4 and Figure 4 report Equations (10) and (11). The random forest has $\Delta = -0.025$: its macro-average external AUC *exceeds* its macro-average internal AUC. Read naively, this says that the random forest performs better on hospitals it has never seen than on the hospital it was trained on, which is not a coherent claim about robustness. The correct reading is that the two averages are taken over different populations: the internal average is dragged down by Switzerland's near-degenerate within-site cross-validation (0.647) and VA's (0.648), whereas the external average includes many transfers *into* the two well-behaved cohorts.

The MLP illustrates the complementary trap. Its gap is the smallest in magnitude (+0.003), which under a $|\Delta|$-minimising selection rule would make it the "most robust" model. But its absolute external AUC is the lowest of the three (0.709). A model that is uniformly mediocre everywhere has a small gap by construction. Gap minimisation and utility maximisation are different objectives, and only the latter is clinically meaningful.

5 DISCUSSION

5.1 What the experiment supports

(1) Internal discrimination on Cleveland with all thirteen variables is strong (AUC = 0.910).
(2) Transport across these four cohorts is heterogeneous rather than uniformly degrading; the directed matrix is strongly asymmetric.

(3) Under $\mathcal{F}_9$ the random forest achieves the highest macro-average external AUC in this run, and is markedly more tolerant of a degenerate source cohort than the linear model.
(4) Source-cohort quality—size, prevalence balance, and site-specific missingness—explains more variance in transfer performance than model family does.

5.2 What it does not support

The study reports no confidence intervals, no calibration curves [10], no threshold-specific sensitivity and specificity, no subgroup audit, and no decision-curve analysis [11]. Four historical cohorts recorded in the late 1980s do not establish contemporary clinical generalization, and the TRIPOD reporting criteria for a deployable model are not met [2]. Macro-averaging weights every directed pair equally, which ignores that target cohorts differ in size by a factor of 2.5 and that pairs sharing a source are statistically dependent. With twelve directed transfers derived from four sites, the effective number of independent external observations is four, not twelve.

5.3 Implications for reporting

Three recommendations follow directly from Figure 2.

First, report the matrix. A single external AUC is a projection of a $|\mathcal{S}|\times|\mathcal{S}|$ object onto one scalar, and the projection destroys exactly the asymmetry that a deploying institution needs to see.

Second, report absolute target-site performance alongside any gap statistic. The quantity a hospital cares about is $\mathrm{AUC}_{s\to t}$ for its own t, not Δ.

Third, audit the recording process before choosing a shared feature set. The Swiss cholesterol column is a visible instance of a general hazard: a variable that is present in the schema but absent in the data will be silently repaired by any pooled imputation step, converting a missing-data problem into an invisible leakage problem.

5.4 Threats to validity

The diagonal and off-diagonal entries of Figure 2 are estimated by different procedures (within-site cross-validation versus a single source-only fit), so they carry different variance. The MLP is sensitive to library version and hardware because early stopping interacts with thread scheduling; qualitative conclusions should be drawn from the matrix structure and not from a third decimal place. Finally, the binarisation of Equation (1) discards severity information that a graded outcome would retain.

6 CONCLUSION

External validity is a matrix of source–target behaviours, not a badge earned from one held-out split. On the four UCI heart-disease cohorts, a model with AUC = 0.910 under internal validation transports at anywhere between 0.571 and 0.888 depending on which pair of hospitals is considered, and the scalar generalization gap is capable of ranking the weakest model first. We recommend that multi-site clinical machine-learning studies report the full directed transfer matrix, report absolute target performance, and treat the source cohort's recording quality as a first-class experimental variable.

DATA AND CODE AVAILABILITY

The cohorts are the four public UCI Heart Disease files named `processed.site.data`, where *site* ranges over `cleveland`, `hungarian`, `switzerland`, and `va`. The analysis notebook, the exported pairwise score table, and the figure-generation script accompanying this manuscript reproduce every number reported here.

ETHICAL STATEMENT

This work uses a de-identified public benchmark. It is a methodological study of external validation and is explicitly not a diagnostic device. No clinical decision should be derived from any model reported here.

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